CREDIT DEFAULT COMPASS: NAVIGATING RISK WITH MACHINE LEARNING

EXECUTIVE SUMMARY

In today's competitive credit landscape, minimizing losses from defaults while avoiding unnecessary outreach is critical – and inherently involves a trade-off between recall (catching true defaulters) and precision (limiting false alarms). We developed and compared multiple machine-learning models (Logistic Regression and Random Forest), then addressed the recall-precision trade-off by first optimizing thresholds for high recall with controlled precision (F1-optimal and recall targeted strategies), and subsequently applying a cost-sensitive threshold under a 10:1 missed-default to false-alarm cost ratio to minimize expected business loss. Using the UCI dataset of 30,000 clients, we engineered interpretable features that capture payment behaviour, credit utilization, and delay patterns, and trained a tuned Random Forest that achieved a test-set AUC of 0.774. By comparing F1-optimal, recall-targeted, and cost-sensitive cutoffs, we demonstrate how each balances detection rate and outreach cost, and recommend the cost-minimizing threshold that best aligns with our pilot cost assumption.

INTRODUCTION & OBJECTIVE:

1. **F1-Optimal Thresholding**: Select the cutoff that maximizes the harmonic mean of precision and recall, balancing false positives and false negatives.
2. **Recall-Targeted Thresholding**: Set a minimum recall (e.g., 60%) to capture a predefined proportion of true defaulters while monitoring precision.
3. **Cost-Sensitive Thresholding**: Perform a cost-based analysis – assuming a pilot ratio of 10:1 for missed defaulters vs. false-alarm costs – to identify the cutoff that minimizes expected business loss.

Credit-card portfolios face two main risks: uncollected losses from defaulters and wasted resources on false alarms. Our approach uses a three-pronged theshold to balance these objectives

Key Insight: This three-pronged approach ensures we not only tune for model performance (F1 & recall) but also align the final decision boundary with cost constraints and strategic objectives.

DATA PREPARATION: The UCI Credit

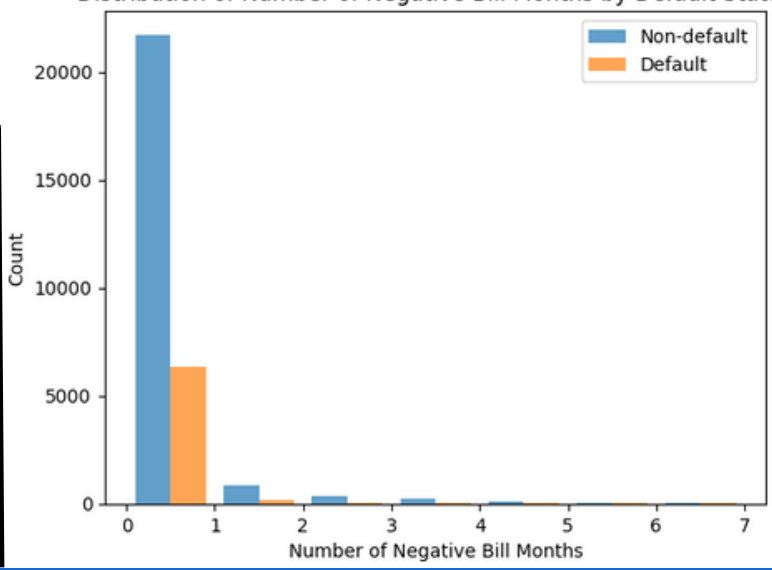
Card Clients dataset (30,000 records, 25 features) underwent the following steps to ensure data integrity and consistency:

- **Category recoding**: Combined miscoded values in EDUCATION and MARRIAGE into an "Other" group.
- **Outlier management:** Winsorized BILL_AMT and PAY_AMT extremes at the 1st – 99th percentiles.
- **Behavioural flag:** Counted months with negative bill amounts (num_neg_bill_months) as a proxy for over-payment discipline.

Table 1 - Cleaning summary

	Operation	Value
0	Rows before cleaning	30000
1	Rows after cleaning	29965
2	Invalid EDUCATION recoded	345
3	Invalid MARRIAGE recoded	54
4	Duplicates dropped	0
5	Negative-bill months flagged	3932
6	Winsorization applied	1st–99th percentile clip

Fig 1 Distribution of Number of Negative Bill Months by Default Status



FEATURE ENGINEERING:

We derived features that encapsulates key risk drivers:

- **Amount Aggregates:** avg_bill, avg_pay to smooth over monthly volatility.
- **Utilization & payment ratios:** max_delay, avg_delay, ever_late + categorical delay_bucket (on_time, short/medium/long delay).
- **Trends:** bill_trend and pay_trend to capture momentum in outstanding balances and payments.

Fig 2

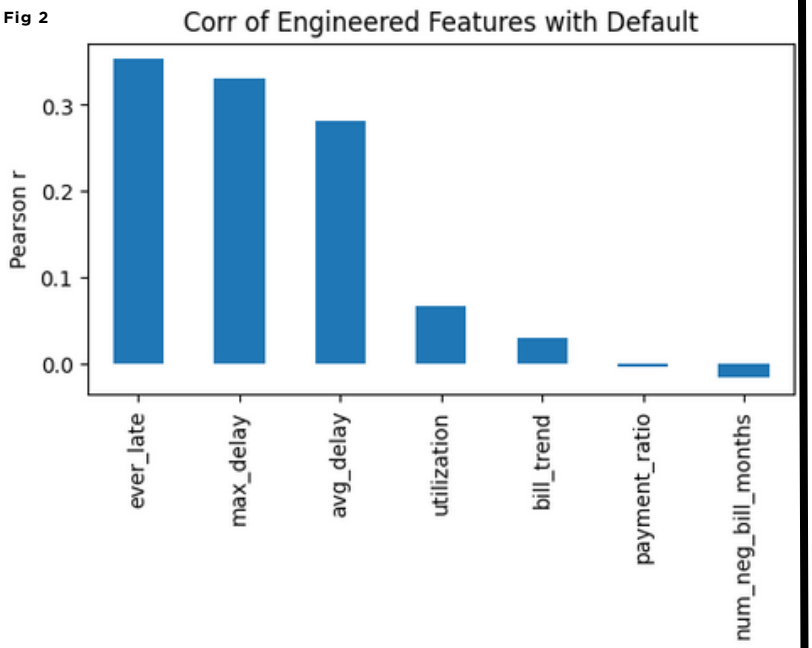
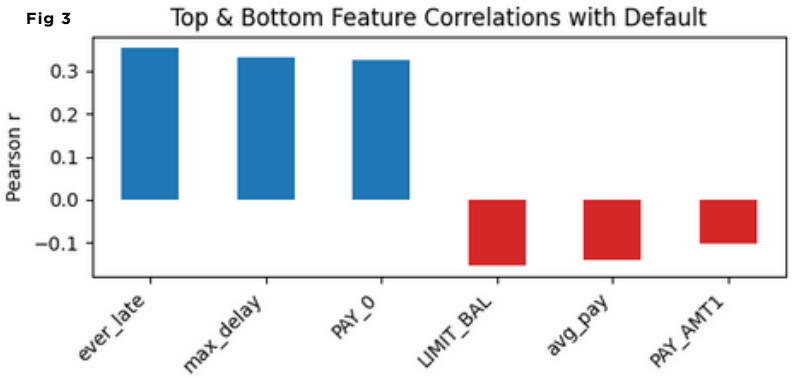


Fig 3



MODELING & SELECTION:

Two pipelines were assessed:

- **Logistic Regression:** with L1 regularization (best AUC = 0.763).
- **Random Forest**: Tuned via GridSearchCV (n_estimators = 200, max_depth = 10, min_samples_leaf = 5, max_features = 'sqrt'), delivering AUC = 0.786 (CV) and AUC = 0.774 (test).

Table 2

```
Tuned Logistic Regression — Classification Report
              precision    recall  f1-score   support

           0       0.88      0.79      0.83      4667
           1       0.45      0.62      0.52      1326

    accuracy                           0.75      5993
   macro avg       0.67      0.70      0.68      5993
weighted avg       0.78      0.75      0.76      5993

Tuned Logistic Regression — ROC AUC: 0.755

Tuned Random Forest — Classification Report
              precision    recall  f1-score   support

           0       0.88      0.83      0.85      4667
           1       0.49      0.59      0.54      1326

    accuracy                           0.77      5993
   macro avg       0.68      0.71      0.69      5993
weighted avg       0.79      0.77      0.78      5993

Tuned Random Forest — ROC AUC: 0.774
```

Key Insight: Based on higher cross-validated and test-set AUC (0.786 CV, 0.774 test) and stronger F1/precision-recall balance, the Random Forest was selected as the final model over Logistic Regression for deployment.

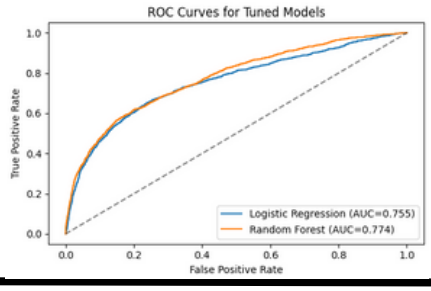


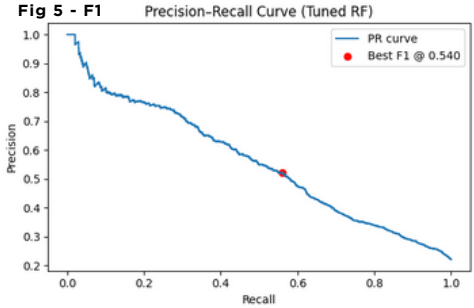
Fig 4

THRESHOLD STRATEGIES:

To align with business goals, we explore three cutoff methods.

- **F1-Optimal (thr = 0.54):** Balanced metrics (Precision = 0.52, Recall = 0.56)
- **Recall-Targeted (60% recall @ thr = 0.479:** Captures 60% of defaulters at 48% precision (FPR = 18.8%).
- **Cost-Optimal (10:1 ratio @ thr = 0.196:** Minimizes expected cost (Precision = 0.26, Recall = 0.96, avg cost = 0.70/account).

Fig 5 - F1



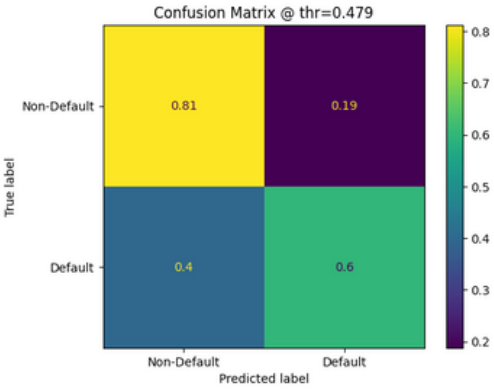


Fig 6 - Recall Targeted Confusion Matrix

Fig 7 - Cost-Optimal Model (Confusion Matrix

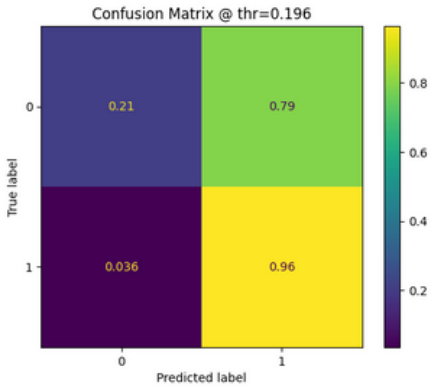


Table 3 - Summary of Threshold Strategies and Key Metrics

	Strategy	Threshold	Precision	Recall	F1	FPR
0	F1-optimal	0.540	0.520	0.56	0.54	0.210
1	Recall-targeted	0.479	0.475	0.60	0.53	0.188
2	Cost-optimal (10:1)	0.196	0.260	0.96	0.41	0.790

NEXT STEPS & BUSINESS CONSIDERATIONS

- **Refine cost assumptions:** Update cost_FN and cost_FP using actual LGDxEAD and true outreach costs.
- **Calibrate probabilities:** Employ CalibratedClassifierCV to produce well-calibrated probability estimates.
- **Deploy & Monitor:** Integrate the final model and threshold into a production scoring pipeline; track live recall, precision, and false-alarm rates, schedule retraining as performance drifts.

FEATURE IMPORTANCE:

- **PAY_0** (most recent delay) and **max_delay** dominate, highlighting the critical role of last-month behaviour.
- Flags like **ever_late** and **delay_bucket** underscore that any history of lateness significantly raise default risk.
- **Utilization** and **avg_delay** also contribute meaningfully.

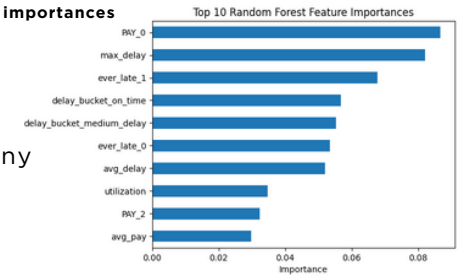



Fig 8 - Feature importances